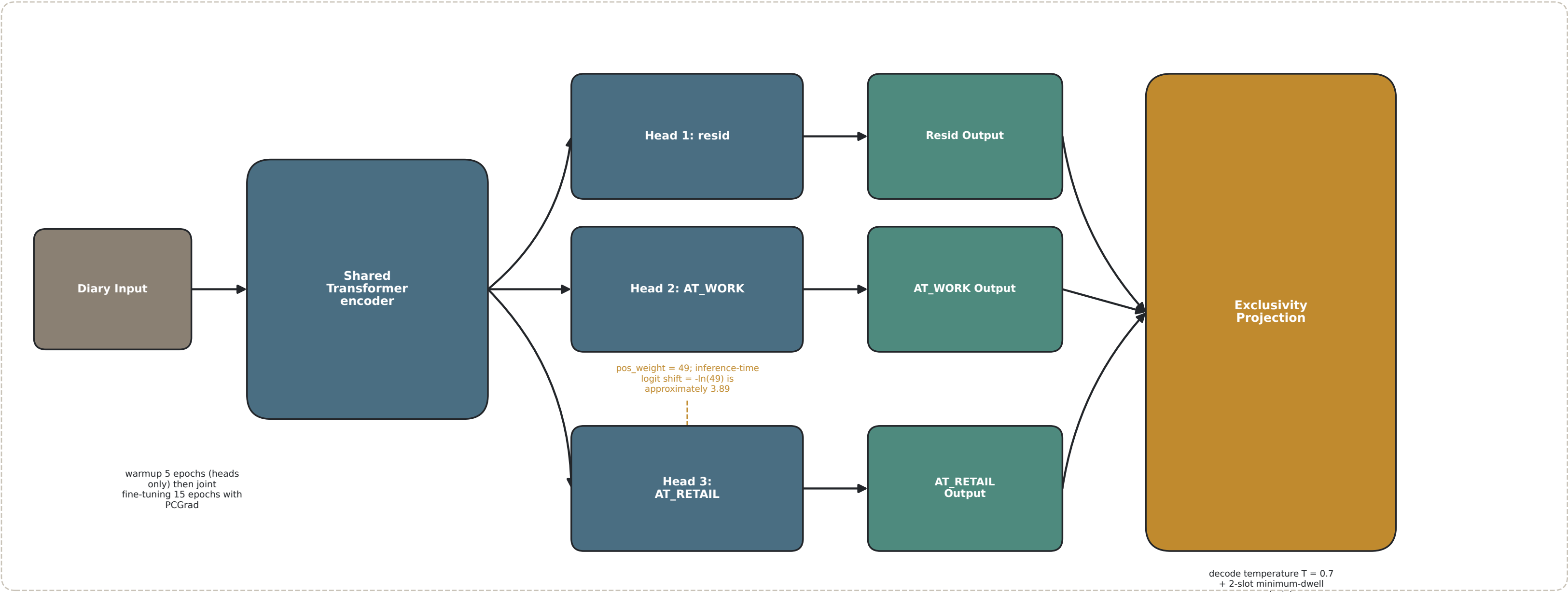


3 GSS heads + 1 non-GSS side-track



warmup 5 epochs (heads only) then joint fine-tuning 15 epochs with PCGrad

pos_weight = 49; inference-time logit shift = -ln(49) is approximately 3.89

loss weights a_resid : a_work : a_retail = 1.0 : 0.5 : 0.3

fixed-weight scalarization + PCGrad (SLAW/UW dropped, unstable on the ~2% AT_RETAIL head)

decode temperature T = 0.7 + 2-slot minimum-dwell constraint

decode thresholds 0.50 / 0.40 / 0.15 (resid / AT_WORK / AT_RETAIL)



SARIMA(1,1,1)(1,1,1,12) per province; population-aggregate monthly series, no GSS respondents behind it -- bypasses the Transformer entirely